IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

**ADVANCED ELECTRONIC DEVICES**

Tuesday, 28 April 2026, 10:00 AM

Time allowed: 3:00 hours

**There are FIVE questions on this paper.**

**Answer Question ONE and THREE other questions.**

*Question One carries 40 marks, and all other questions carry 20 marks each.*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : K. Fobelets
Second Marker(s) : Z. Durrani

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## Constants

| | |
|---|---|
| permittivity of free space: | $\varepsilon_o = 8.85 \times 10^{-12}$ F/m |
| permeability of free space: | $\mu_o = 4\pi \times 10^{-7}$ H/m |
| charge of an electron: | $e = 1.6 \times 10^{-19}$ C (1 eV) |
| thermal voltage: | $kT/e = 0.026$V at $T = 300$K |
| Planck constant | $h = 6.626 \times 10^{-34}$ J.s |
| Boltzmann constant | $k = 1.38 \times 10^{-23}$ J/K |
| Electron rest mass | $m = 0.911 \times 10^{-30}$ kg |
| Speed of light | $c = 3 \times 10^{8}$ m/s |
| intrinsic carrier concentration in Si: | $n_i = 1.45 \times 10^{10}$ cm$^{-3}$ at $T = 300$K |
| intrinsic carrier concentration in Ge: | $n_i = 2.4 \times 10^{13}$ cm$^{-3}$ at $T = 300$K |
| dielectric constant of Si: | $\varepsilon_r = 12$ |
| dielectric constant of Ge: | $\varepsilon_r = 16$ |
| dielectric constant of $SiO_2$: | $\varepsilon_{ox} = 4$ |
| electron affinity of Si: | $\chi = 4.05$ eV |
| electron affinity of Ge: | $\chi = 4.00$ eV |
| effective density of states of Si: | $N_C = 2.8 \times 10^{19}$ cm$^{-3}$<br>$N_V = 1.04 \times 10^{19}$ cm$^{-3}$ |
| effective density of states of Ge: | $N_C = 1.04 \times 10^{19}$ cm$^{-3}$<br>$N_V = 6.0 \times 10^{18}$ cm$^{-3}$ |
| lattice constant of Si: | $a = 0.543$ nm |
| lattice constant of Ge: | $a = 0.565$ nm |
| lattice constant of SiC: | $a = 0.307$ nm |
| bandgap of Si: | $E_G = 1.12$ eV |
| bandgap of Ge: | $E_G = 0.66$ eV |
| thermal conductivity Si: | $\kappa = 1.5$ W/cm C |
| thermal conductivity $SiO_2$: | $\kappa = 0.6$ W/cm C |

*Table 0: Workfunction of metals*

| metal | $\phi$ (V) |
|---|---|
| Al | 4.1 |
| Ni | 5.0 |
| Ti | 4.3 |
| W | 4.6 |

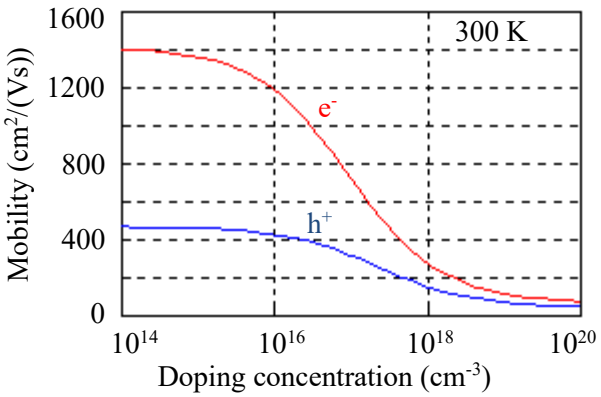



*Figure 0: Electron and hole mobility in Si as a function of doping concentration.*

## Formulae

$p = h/\lambda = \hbar k$ — De Broglie equations

$E = h\nu = \hbar\omega$

$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi = i\hbar\frac{\partial\psi}{\partial t}$ — Schrödinger equation in one-dimension

$p = N_v e^{(E_v - E_F)/kT}$ — Carrier density – Fermi-Dirac statistics

$n = N_c e^{(E_F - E_c)/kT}$

$J_n(x) = e\mu_n n(x)E(x) + eD_n\frac{dn(x)}{dx}$ — Drift – diffusion current density equations for electrons, $J_n$ and holes, $J_p$

$J_p(x) = e\mu_p p(x)E(x) - eD_p\frac{dp(x)}{dx}$

$I_{DS} = \frac{\mu C_{ox} W}{L}\left((V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2}\right)$ — Strong inversion MOSFET current

$V_{th} = \phi_m - \phi_s + 2\times\phi_F + \gamma\times\sqrt{2\times\phi_F}$ — Threshold voltage

$\phi_F = \frac{kT}{e}\ln\left(\frac{N_A}{n_i}\right)$ — Fermi potential (difference between intrinsic and Fermi level)

$\gamma = \frac{\sqrt{2e\varepsilon_s N_A}}{C_{ox}}$ — Body effect coefficient

$J = \frac{eD_n n_p}{L_n}\left(e^{\frac{eV}{kT}} - 1\right) + \frac{eD_p p_n}{L_p}\left(e^{\frac{eV}{kT}} - 1\right)$ — Diode diffusion current density

$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_A N_D}{n_i^2}\right)$ — Built-in voltage pn diode

$W_{depl}(V) = \left[\frac{2\varepsilon(V_{bi} - V)}{e}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\right]^{1/2}$ — Depletion width in pn diode

$S = \frac{dV_{GS}}{dLog(I_{DS})}$ — Sub-threshold swing

$c = c_0\exp\left(\frac{eV}{kT}\right)$ with $\begin{cases} c = p_n \text{ or } n_p \\ c_0 \text{ bulk minority carrier concentration} \end{cases}$ — Minority carrier injection under bias $V$

$\delta c = \Delta c\exp\left(\frac{-x}{L}\right)$ with $\begin{cases} \delta c = \delta p_n \text{ or } \delta n_p \\ \Delta c \text{ the excess carrier concentration} \\ \text{at the edge of the depletion region} \end{cases}$ — Excess carrier concentration as a function of distance

$L = \sqrt{D\tau}$ — Diffusion length

$D = \frac{kT}{e}\mu$ — Einstein relation

$J = \sigma\frac{dV}{dx} + eD\frac{dn}{dx} - \sigma S\frac{dT}{dx}$ — 1D current density with thermal gradient

$J_Q = \frac{J^2}{\sigma} + STJ - \kappa\frac{dT}{dx}$ — 1D heat flux

1. Compulsory question.

a) A long-channel Si MOSFET with a $SiO_2$ gate oxide and a non-depletable polySi gate electrode is made on a substrate with a doping density of $N_A = 10^{17}$ cm$^{-3}$. When extracting the transconductance, $g_m$ from experiments in the triode region (low $V_{DS}$) at room temperature, three observations are made:

1. $g_m$ is not constant as a function of gate voltage, $V_{GS}$.
2. The maximum transconductance gives a mobility value of $\mu = 450$ cm$^2$/Vs.
3. The threshold voltage, $V_{th}$ is lower than that expected from the theoretical calculations.

Answer the following questions.

(i) Is this an n- or p-MOS? [1]

(ii) What caused $V_{th}$ to be lower than theoretically expected (see formulae sheet)? Plot an energy band diagram to illustrate your answer. [4]

(iii) What causes $g_m$ to decrease for higher gate voltages? Choose from: phonon scattering, ionised impurity scattering, Coulomb scattering, surface roughness scattering. Multiple scattering processes may be responsible. [2]

(iv) Why is the maximum value $\mu$ lower than what is reported in Fig. 0 on p.1? Choose from: phonon scattering, ionised impurity scattering, Coulomb scattering, surface roughness scattering and/or parasitic resistances. Multiple factors may be responsible. [3]

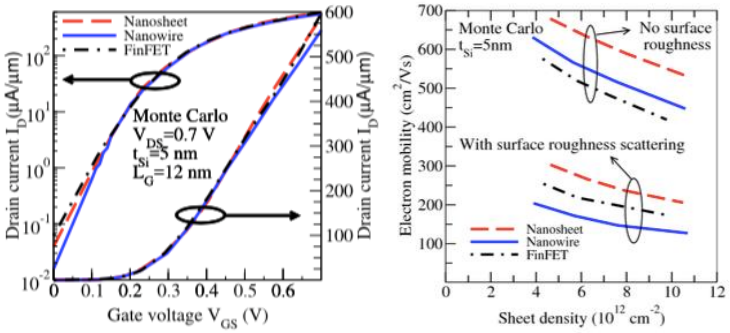
b) A midgap gate metal has a workfunction, $\phi_m$ that is equal to the workfunction of undoped Si, $\phi_{Si}$.

(i) Draw the energy band diagram from gate through oxide into the channel of both an nMOS and pMOS with the same substrate doping density and oxide thickness, using a midgap gate metal in both cases. [6]

(ii) Write the relationship between the threshold voltage, $V_{th}$ of the n and p-MOS. [2]

(iii) How can a gate electrode be made with an approximately midgap workfunction? [2]

c) The figure below gives the transfer characteristics of three MOSFETs with different geometries. To analyse the reasons for the difference in characteristics, TCAD Sentaurus simulations are carried out. The results of these simulations are also shown in the figure.




*Left: transfer characteristics and right: TCAD simulations*

TURN OVER

(i) What is the cause of surface roughness scattering in MOSFETs. [2]

(ii) The part of the Physics section in sDevice related to mobility is:

| Line | Command |
|---|---|
| 1 | Physics(Material="Silicon"){ |
| 2 | Mobility( |
| 3 | PhuMob |
| 4 | hHighFieldSaturation(GradQuasiFermi) |
| 5 | eHighFieldSaturation(CarrierTempDrive) |
| 6 | Enormal |
| 7 | DopingDependence |
| 8 | ) } |

Which of these commands will need to be removed to simulate the current-voltage characteristics without surface roughness scattering? [2]

(iii) Based on the current-voltage characteristics in the figure, comment on the performance parameters, $g_m$, SS and $I_{on}/I_{off}$, of the different implementations. [6]

d) Assuming the wave function of an electron of mass $m$ travelling in a potential $V$ is given by $\Psi = A.\exp[i(kx - \omega t)]$, derive the 1-D time-dependent form of Schrödinger's equation.

[4]

e) Calculate the de Broglie wavelength for the following:

(i) An electron of kinetic energy 5 eV.

[3]

(ii) A photon of energy 5 eV.

[3]

2. MOSFETs

a) A short-channel Si n-MOS with a $HfO_2$ gate insulator and a metal gate electrode shows drain-induced barrier lowering DIBL = 160 mV/V and a sub-threshold swing SS = 120 mV/dec.

(i) Sketch appropriate current-voltage characteristics from which both DIBL and SS can be derived and define the values of DIBL and SS on this plot. [4]

(ii) Sketch an energy band diagram to explain the physical phenomenon causing DIBL. [4]

(iii) Why must a thin $SiO_2$ layer be thermally grown on Si before deposition of $HfO_2$? [2]

b) From the 90 nm node onwards, strain is used in MOSFETs to enhance performance without gate length reduction. To implement strain, heterojunctions of Si and Si alloys are made, mainly using C or Ge. Local strain is the commercially implemented method.

(i) Sketch the material cross-section of a planar Si n-MOSFET in which local strain is implemented. Define the material and doping in all regions of your sketch. [4]

(ii) What type of strain, tensile or compressive, is implemented in (i)? [2]

(iii) Why does this strain implementation improve performance? Give the main parameter that varies and the reasons why that variation happens. Also, give the additional condition for this parameter improvement to be useful in a MOSFET implementation. [4]

3. finFETs

a) Sketch a material cross-section through the fin in a plane perpendicular to the current flow from source to drain for both a finFET on SOI and a finFET on bulk using the STI (shallow trench isolation) process. Identify the main difference in structure between these two approaches. Give one strength and one weakness for each implementation. [8]

b) The definition of the subthreshold slope is given by:

$$SS = 2.3\,\frac{kT}{e}\left(1 + \frac{C_{depl} + C_{it}}{C_{ox}}\right)$$

With $k$ Boltzmann constant, $T$ absolute temperature, $e$ electron charge value, $C_{depl}$, $C_{it}$, and $C_{ox}$, the depletion, interface traps and oxide capacitance per area, respectively.

(i) Why can $C_{depl}$ be neglected in finFETs? [2]
(ii) How can $C_{it}$ be minimised? [2]
(iii) If the interface charge density is zero, what is the best SS value that can be achieved in the finFET at room temperature? [2]

c) An ultra-thin finFET is made with a low doping density in the fin. Above threshold, $V_{GS} > V_{th}$, volume inversion happens.

(i) What is the electrostatic cause for this volume inversion? [2]
(ii) Sketch an energy band diagram through the fin that illustrates this effect. [4]

4. SOI

a) Sketch the material cross-section of a CMOS on bulk and on SOI. Give one advantage and one weakness of CMOS on SOI and identify them in your cross-section. [8]

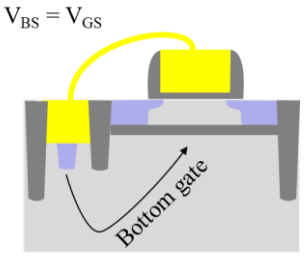
b) In partially depleted SOI, the threshold voltage can suddenly shift to lower values for the MOSFET working in its saturation region. Illustrate why. [4]

c) FDSOI

(i) Derive an approximate expression for the threshold voltage, $V_{th,}$ for a fully depleted SOI nMOS in the case of a thick BOX layer and grounded bulk. Explain any assumptions you make. [4]

(ii) Explain why this approximation cannot be used in UTBB-SOI (ultra-thin body and box SOI). [2]

d) The figure below gives a sketch of the material cross section of an UTBB-SOI n-MOS with front and back gate connected. If the threshold voltage is $V_{th0}$ at $V_{GS} = 0$ V, will the threshold voltage $V_{thD}$ at $V_{GS} = V_{DD}$ be smaller or larger than $V_{th0}$? Explain briefly why. [2]




*DT mode connected UTBB-SOI n-MOS.*

5. Quantum devices

An electron travelling in the positive $x$-direction is incident from the region $x < 0$ onto a one-dimensional potential barrier of height $V_B$ and width $L$. The potential energy $V(x)$ is given by:

$V(x) = V_B$, for $0 \leq x \leq L$
$V(x) = 0$, elsewhere

a) Write down the general form of the time-independent wavefunctions $\psi(x)$ when $x < 0$, $x$ lies between 0 and $L$, and when $x > L$. [3]

b) Write down expressions for the wave-vectors when $x < 0$, $x$ lies between 0 and $L$, and $x > L$. [6]

c) Using the wavefunctions $\psi(x)$ in the different regions, and the boundary conditions at $x = 0$ and $x = L$, write down equations linking the wavefunctions at $x = 0$ and $x = L$. [5]

d) Hence, derive an expression for the amplitude transmission coefficient $t$ for the electron travelling across the barrier. [6]